Algebra II Final Practice: Conics, Statistics, Probability, and Functions

Instructions

Complete in 60 minutes. Show clear work. Use exact forms for conic features unless a decimal approximation is explicitly requested. For statistics, use population standard deviation and round it to the nearest hundredth.

Reference Facts

- Circle:
$$(x-h)^2 + (y-k)^2 = r^2$$
- Parabolas:
$$(x-h)^2 = 4p(y-k), \qquad (y-k)^2 = 4p(x-h)$$
- Ellipses:
$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, \qquad c^2 = a^2 - b^2$$
- Hyperbolas:
$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1, \qquad \frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1, \qquad c^2 = a^2 + b^2$$
- Z-score:
$$z = \frac{x-\mu}{\sigma}, \qquad x = \mu + z\sigma$$
- Counting:
$${}_nC_r = \frac{n!}{r!(n-r)!}$$

Problems

1. Identify each conic as a circle, parabola, ellipse, or hyperbola. Do not rewrite in standard form.
$$x^2 + y^2 - 6x + 8y - 11 = 0$$
$$y^2 - 4y - 12x + 16 = 0$$
$$9x^2 + 4y^2 - 54x + 16y + 61 = 0$$
$$16y^2 - 25x^2 - 96y - 50x - 281 = 0$$

2. Rewrite the circle in standard form. Then identify its center and radius.
$$x^2 + y^2 + 10x - 12y + 25 = 0$$
Then write the equation of the circle whose diameter has endpoints $(-2, 7)$ and $(8, -1)$.

3. Rewrite the parabola in standard form. Then state its orientation, vertex, focus, directrix, and axis of symmetry.
$$x^2 + 8x - 12y + 4 = 0$$

4. For the ellipse
$$\frac{(x-2)^2}{25} + \frac{(y+1)^2}{9} = 1,$$
state the center, orientation, vertices, co-vertices, and foci.

5. Write an equation of the ellipse with center $(1, -3)$, vertices $(1, 5)$ and $(1, -11)$, and foci $(1, 2)$ and $(1, -8)$.

6. For the hyperbola
$$\frac{(y-4)^2}{16} - \frac{(x+2)^2}{9} = 1,$$
state the center, orientation, vertices, co-vertices, foci, and asymptotes.

7. Write an equation of the hyperbola with center $(3, -2)$, vertices $(-2, -2)$ and $(8, -2)$, and foci $(-4, -2)$ and $(10, -2)$. Then state its asymptotes.

8. The data set below gives quiz scores.
$$42,\ 45,\ 47,\ 47,\ 50,\ 52,\ 54,\ 57,\ 61,\ 65$$
Find the mean, median, mode, range, population standard deviation, five-number summary, and IQR.

9. A set of final exam scores is approximately normal with mean 76 and standard deviation 8.
   (a) Find the z-score for a score of 90.
   (b) Find the exam score with $z = -1.25$.
   (c) Interpret the meaning of the z-score from part (a) in a complete sentence.

10. A club has 8 juniors and 6 seniors. A committee of 4 students is chosen at random.
    (a) How many committees are possible?
    (b) How many committees have exactly 2 juniors and 2 seniors?
    (c) What is the probability that the committee has exactly 2 juniors and 2 seniors?

11. A parking permit code uses 2 different uppercase letters followed by 3 digits. The first digit cannot be 0, and digits may repeat.
    (a) How many permit codes are possible?
    (b) How many permit codes are possible if the digits may not repeat?

12. A theater sells 300 tickets when the price is \$12. For each \$1 increase in price, the theater expects to sell 15 fewer tickets. Let x be the number of \$1 price increases.
    (a) Create a price function $p(x)$ and a ticket-sales function $t(x)$.
    (b) Create a revenue function $R(x)$.
    (c) Find the ticket price that maximizes revenue.
    (d) Find the maximum revenue and the number of tickets sold at that price.
    (e) State a reasonable practical domain for x.